EE414 — Numerical Methods Assignment

First-Order RL Circuit: Analytic Solution and Numerical ODE Solvers

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Q1 — Derivation of the ODE model for y = vR

The step-excited series RL circuit supplies a 1 V source at t = 0. Applying Kirchhoff's Voltage Law around the loop gives

$$1\ \mathrm{V} = v_L + v_R$$

The same current i flows through both elements, so vR = iR and the resistor voltage is *y = iR*. Rearranging gives i = y/R, which lets vL be written in terms of y:

$$v_R = y = iR, \qquad v_L = L\frac{di}{dt} = \frac{L}{R}\frac{dy}{dt}$$

Substituting into KVL and dividing through by R:

$$1 = \frac{L}{R}\frac{dy}{dt} + y \implies \frac{dy}{dt} = \frac{1}{\tau}(1-y), \quad \tau = \frac{L}{R}$$

which is the required first-order linear ODE:

$$\frac{dy}{dt} = \frac{1}{\tau}(1-y), \qquad \tau = \frac{L}{R}$$

For the assigned parameters R = 4 MΩ and L = 580 µH, the time constant is

$$\tau = \frac{L}{R} = \frac{580\ \mu\mathrm{H}}{4\ \mathrm{M\Omega}} = 145\ \mathrm{ps}$$

Time is expressed in picoseconds throughout the assignment, so τ = 145 (no trailing units inside the update formulas). The initial condition is y(0) = 0 because the inductor current — and hence vR — is zero immediately before the step.

Q2 — Analytic solution (elegant substitution)

The initial-value problem is

$$\frac{dy}{dt} = \frac{1-y}{\tau}, \qquad y(0) = 0$$

The right-hand side depends only on the quantity (1 − y) — the deviation of y from its equilibrium value 1. It is therefore natural to track that deviation directly. Let *u(t) = 1 − y(t)*:

$$u(t) = 1 - y(t) \implies \dot{u} = -\dot{y}$$

and the ODE becomes the canonical exponential-decay equation

$$\frac{du}{dt} = -\frac{u}{\tau}$$

whose general solution is

$$u(t) = Ce^{-t/\tau}, \qquad u(0) = 1 \Rightarrow C = 1$$

Back-substituting u = 1 − y gives the analytic solution:

$$y(t) = 1 - e^{-t/\tau}, \qquad \tau = \frac{L}{R}$$

Verification. At t = 0, y(0) = 1 − 1 = 0 ✓. Differentiating, dy/dt = (1/τ) e^(−t/τ) = (1 − y)/τ ✓.

The physical reading is clean: y relaxes from 0 toward the steady-state value 1 with a single time constant τ = 145 ps. At t = τ it has covered $1 - e^{-1} \approx 63.2\%$ of the gap, and at 5τ more than 99.3% — the standard first-order RL step response.

This substitution route sidesteps the logarithm integral that separation of variables would introduce: one change of variable reveals that the error from steady state is what actually obeys pure exponential decay, and the answer falls out by inspection.

Numerical solvers (Q3 – Q5)

All three methods approximate the solution on the grid $t_i = i \cdot h$ for i = 0, 1, ..., N where N·h = 240 ps. The right-hand side is

$$f(t, y) = \frac{1-y}{\tau} = \frac{1-y}{145} \quad [\mathrm{ps}^{-1}]$$

The exact solution $y(t_i)$ from Q2 is used to evaluate the truncation error

$$E_t = |\, y(t_i) - \hat{y}(t_i) \,|$$

In the tables below, $\hat{y}(t_i)$ is the numerical estimate, $y(t_i)$ is the analytic value from Q2, and Et is tabulated to two significant figures in scientific notation. All values ŷ and y are shown to five significant figures. The driver code (common to all three methods) is:

```
% EE414 – Individual parameters: R = 4 MΩ, L = 580 µH  =>  tau = 145 ps
R = 4e6; L = 580e-6;
tau = (L/R)*1e12;            % tau in picoseconds (= 145)

f       = @(t,y) (1 - y)/tau;   % ODE right-hand side (t in ps)
y_exact = @(t)    1 - exp(-t/tau);
t_end   = 240;                  % simulate from 0 to 240 ps
hVec    = [40, 20, 10];         % step sizes
```

Q3 — Euler's method

Euler advances the solution using the slope at the start of each interval:

$$y_{i+1} = y_i + h\, f(t_i,\, y_i)$$

It is the simplest explicit method but also the least accurate — global error is O(h). For this IVP the method always overshoots, because f decreases as y grows, so the start-of-interval slope over-estimates the mean slope across the step. The implementation is the single line inside the for-loop below.

```
function [t,y] = euler(f, t0, y0, h, tEnd)
    N = round((tEnd - t0)/h);
    t = (t0:h:tEnd).';
    y = zeros(N+1, 1);  y(1) = y0;
    for i = 1:N
```

```
        y(i+1) = y(i) + h*f(t(i), y(i));
    end
end
```

Results for h = 40, 20, 10 ps:

h = 40 ps

i	t_i	$\hat{y}(t_i)$	$y(t_i)$	E_t
0	0	0.00000	0.00000	0
1	40	0.27586	0.24108	3.5×10^{-2}
2	80	0.47562	0.42404	5.2×10^{-2}
3	120	0.62028	0.56290	5.7×10^{-2}
4	160	0.72503	0.66827	5.7×10^{-2}
5	200	0.80088	0.74825	5.3×10^{-2}
6	240	0.85581	0.80894	4.7×10^{-2}

h = 20 ps

i	t_i	$\hat{y}(t_i)$	$y(t_i)$	E_t
0	0	0.00000	0.00000	0
1	20	0.13793	0.12884	9.1×10^{-3}
2	40	0.25684	0.24108	1.6×10^{-2}
3	60	0.35934	0.33886	2.0×10^{-2}
4	80	0.44771	0.42404	2.4×10^{-2}
5	100	0.52389	0.49825	2.6×10^{-2}
6	120	0.58956	0.56290	2.7×10^{-2}
7	140	0.64617	0.61921	2.7×10^{-2}
8	160	0.69497	0.66827	2.7×10^{-2}
9	180	0.73705	0.71101	2.6×10^{-2}
10	200	0.77332	0.74825	2.5×10^{-2}
11	220	0.80458	0.78068	2.4×10^{-2}
12	240	0.83154	0.80894	2.3×10^{-2}

h = 10 ps

i	t_i	$\hat{y}(t_i)$	$y(t_i)$	E_t
0	0	0.00000	0.00000	0
1	10	0.06897	0.06664	2.3×10^{-3}
2	20	0.13317	0.12884	4.3×10^{-3}
3	30	0.19296	0.18690	6.1×10^{-3}
4	40	0.24861	0.24108	7.5×10^{-3}
5	50	0.30043	0.29166	8.8×10^{-3}
6	60	0.34868	0.33886	9.8×10^{-3}
7	70	0.39360	0.38292	1.1×10^{-2}
8	80	0.43542	0.42404	1.1×10^{-2}
9	90	0.47436	0.46243	1.2×10^{-2}
10	100	0.51061	0.49825	1.2×10^{-2}
11	110	0.54436	0.53169	1.3×10^{-2}
12	120	0.57578	0.56290	1.3×10^{-2}
13	130	0.60504	0.59203	1.3×10^{-2}
14	140	0.63228	0.61921	1.3×10^{-2}
15	150	0.65764	0.64459	1.3×10^{-2}
16	160	0.68125	0.66827	1.3×10^{-2}
17	170	0.70323	0.69038	1.3×10^{-2}
18	180	0.72370	0.71101	1.3×10^{-2}
19	190	0.74275	0.73027	1.2×10^{-2}
20	200	0.76049	0.74825	1.2×10^{-2}

i	t_i	$\hat{y}(t_i)$	$y(t_i)$	E_t
21	210	0.77701	0.76502	1.2×10^{-2}
22	220	0.79239	0.78068	1.2×10^{-2}
23	230	0.80671	0.79530	1.1×10^{-2}
24	240	0.82004	0.80894	1.1×10^{-2}

Q4 — Heun's method (classical RK2)

Heun's method averages the slope at the start of the interval with the slope at a predicted end-point:

$$k_1 = f(t_i,\ y_i)$$

$$k_2 = f(t_i + h,\ y_i + h\,k_1)$$

$$y_{i+1} = y_i + \frac{h}{2}\,(k_1 + k_2)$$

Two function evaluations per step, global error $O(h^2)$. For the RL step response the trapezoidal averaging cancels most of Euler's systematic overshoot.

```
function [t,y] = rk2(f, t0, y0, h, tEnd)   % Heun's method
    N = round((tEnd - t0)/h);
    t = (t0:h:tEnd).';
    y = zeros(N+1, 1);  y(1) = y0;
    for i = 1:N
        k1 = f(t(i),       y(i));
        k2 = f(t(i) + h,   y(i) + h*k1);
        y(i+1) = y(i) + (h/2)*(k1 + k2);
    end
end
```

Results for h = 40, 20, 10 ps:

h = 40 ps

i	t_i	$\hat{y}(t_i)$	$y(t_i)$	E_t
0	0	0.00000	0.00000	0
1	40	0.23781	0.24108	3.3×10^{-3}
2	80	0.41907	0.42404	5.0×10^{-3}
3	120	0.55722	0.56290	5.7×10^{-3}
4	160	0.66252	0.66827	5.8×10^{-3}
5	200	0.74278	0.74825	5.5×10^{-3}

h = 20 ps

i	t_i	$\hat{y}(t_i)$	$y(t_i)$	E_t
0	0	0.00000	0.00000	0
1	20	0.12842	0.12884	4.2×10^{-4}
2	40	0.24035	0.24108	7.4×10^{-4}
3	60	0.33790	0.33886	9.6×10^{-4}
4	80	0.42293	0.42404	1.1×10^{-3}
5	100	0.49703	0.49825	1.2×10^{-3}

h = 10 ps

i	t_i	$\hat{y}(t_i)$	$y(t_i)$	E_t
0	0	0.00000	0.00000	0
1	10	0.06659	0.06664	5.4×10^{-5}
2	20	0.12874	0.12884	1.0×10^{-4}
3	30	0.18676	0.18690	1.4×10^{-4}
4	40	0.24091	0.24108	1.7×10^{-4}
5	50	0.29145	0.29166	2.0×10^{-4}

i	t_i	$\hat{y}(t_i)$	$y(t_i)$	E_t
6	240	0.80395	0.80894	5.0×10^{-3}

i	t_i	$\hat{y}(t_i)$	$y(t_i)$	E_t
6	120	0.56162	0.56290	1.3×10^{-3}
7	140	0.61792	0.61921	1.3×10^{-3}
8	160	0.66698	0.66827	1.3×10^{-3}
9	180	0.70975	0.71101	1.3×10^{-3}
10	200	0.74702	0.74825	1.2×10^{-3}
11	220	0.77951	0.78068	1.2×10^{-3}
12	240	0.80783	0.80894	1.1×10^{-3}

i	t_i	$\hat{y}(t_i)$	$y(t_i)$	E_t
6	60	0.33863	0.33886	2.3×10^{-4}
7	70	0.38267	0.38292	2.5×10^{-4}
8	80	0.42378	0.42404	2.7×10^{-4}
9	90	0.46215	0.46243	2.8×10^{-4}
10	100	0.49796	0.49825	2.9×10^{-4}
11	110	0.53139	0.53169	3.0×10^{-4}
12	120	0.56259	0.56290	3.0×10^{-4}
13	130	0.59172	0.59203	3.1×10^{-4}
14	140	0.61891	0.61921	3.1×10^{-4}
15	150	0.64428	0.64459	3.1×10^{-4}
16	160	0.66797	0.66827	3.1×10^{-4}
17	170	0.69008	0.69038	3.0×10^{-4}
18	180	0.71072	0.71101	3.0×10^{-4}
19	190	0.72998	0.73027	3.0×10^{-4}
20	200	0.74796	0.74825	2.9×10^{-4}
21	210	0.76474	0.76502	2.8×10^{-4}
22	220	0.78041	0.78068	2.8×10^{-4}
23	230	0.79503	0.79530	2.7×10^{-4}
24	240	0.80868	0.80894	2.6×10^{-4}

Q5 — Classical fourth-order Runge–Kutta (RK4)

RK4 combines four carefully-weighted slope estimates per step:

$$k_1 = f(t_i,\ y_i)$$

$$k_2 = f\left(t_i + \tfrac{h}{2},\ y_i + \tfrac{h}{2}\,k_1\right)$$

$$k_3 = f\left(t_i + \tfrac{h}{2},\ y_i + \tfrac{h}{2}\,k_2\right)$$

$$k_4 = f(t_i + h,\ y_i + h\,k_3)$$

$$y_{i+1} = y_i + \frac{h}{6}\left(k_1 + 2k_2 + 2k_3 + k_4\right)$$

Four function evaluations per step, global error $O(h^4)$. For this smooth, non-stiff IVP the error at h = 10 ps is already at the 10^{-8} level — below any meaningful modelling error from the resistor/inductor tolerances.

```
function [t,y] = rk4(f, t0, y0, h, tEnd)
    N = round((tEnd - t0)/h);
    t = (t0:h:tEnd).';
    y = zeros(N+1, 1);  y(1) = y0;
    for i = 1:N
        k1 = f(t(i),          y(i));
        k2 = f(t(i) + h/2,    y(i) + h*k1/2);
        k3 = f(t(i) + h/2,    y(i) + h*k2/2);
        k4 = f(t(i) + h,      y(i) + h*k3);
        y(i+1) = y(i) + (h/6)*(k1 + 2*k2 + 2*k3 + k4);
    end
end
```

Results for h = 40, 20, 10 ps:

h = 40 ps **h = 20 ps** **h = 10 ps**

i	t_i	$\hat{y}(t_i)$	$y(t_i)$	E_t
0	0	0.00000	0.00000	0
1	40	0.24107	0.24108	1.3×10^{-5}
2	80	0.42402	0.42404	1.9×10^{-5}
3	120	0.56287	0.56290	2.2×10^{-5}
4	160	0.66825	0.66827	2.2×10^{-5}
5	200	0.74823	0.74825	2.1×10^{-5}
6	240	0.80892	0.80894	1.9×10^{-5}

i	t_i	$\hat{y}(t_i)$	$y(t_i)$	E_t
0	0	0.00000	0.00000	0
1	20	0.12884	0.12884	4.1×10^{-7}
2	40	0.24108	0.24108	7.1×10^{-7}
3	60	0.33886	0.33886	9.3×10^{-7}
4	80	0.42404	0.42404	1.1×10^{-6}
5	100	0.49825	0.49825	1.2×10^{-6}
6	120	0.56290	0.56290	1.2×10^{-6}
7	140	0.61921	0.61921	1.2×10^{-6}
8	160	0.66827	0.66827	1.2×10^{-6}
9	180	0.71101	0.71101	1.2×10^{-6}
10	200	0.74825	0.74825	1.2×10^{-6}
11	220	0.78068	0.78068	1.1×10^{-6}
12	240	0.80894	0.80894	1.1×10^{-6}

i	t_i	$\hat{y}(t_i)$	$y(t_i)$	E_t
0	0	0.00000	0.00000	0
1	10	0.06664	0.06664	1.3×10^{-8}
2	20	0.12884	0.12884	2.4×10^{-8}
3	30	0.18690	0.18690	3.4×10^{-8}
4	40	0.24108	0.24108	4.2×10^{-8}
5	50	0.29166	0.29166	4.9×10^{-8}
6	60	0.33886	0.33886	5.5×10^{-8}
7	70	0.38292	0.38292	5.9×10^{-8}
8	80	0.42404	0.42404	6.3×10^{-8}
9	90	0.46243	0.46243	6.7×10^{-8}
10	100	0.49825	0.49825	6.9×10^{-8}
11	110	0.53169	0.53169	7.1×10^{-8}
12	120	0.56290	0.56290	7.2×10^{-8}
13	130	0.59203	0.59203	7.3×10^{-8}
14	140	0.61921	0.61921	7.3×10^{-8}
15	150	0.64459	0.64459	7.3×10^{-8}
16	160	0.66827	0.66827	7.3×10^{-8}
17	170	0.69038	0.69038	7.2×10^{-8}
18	180	0.71101	0.71101	7.2×10^{-8}
19	190	0.73027	0.73027	7.1×10^{-8}
20	200	0.74825	0.74825	6.9×10^{-8}
21	210	0.76502	0.76502	6.8×10^{-8}
22	220	0.78068	0.78068	6.6×10^{-8}
23	230	0.79530	0.79530	6.5×10^{-8}
24	240	0.80894	0.80894	6.3×10^{-8}